

Task 2-Stock trading platform

Source Code

```
import java.util.HashMap;
import java.util.Scanner;

public class StockTradingPlatform {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Simulated stock prices
        HashMap<String, Double> market = new HashMap<>();
        market.put("AAPL", 150.0);
        market.put("GOOGL", 2800.0);
        market.put("AMZN", 3400.0);
        market.put("TSLA", 700.0);

        // Portfolio and balance
        HashMap<String, Integer> portfolio = new HashMap<>();
        double balance = 10000.0; // Starting amount

        boolean running = true;

        while (running) {
            System.out.println("\n--- Stock Trading Platform ---");
            System.out.println("1. View Market Data");
            System.out.println("2. Buy Stock");
            System.out.println("3. Sell Stock");
            System.out.println("4. View Portfolio");
            System.out.println("5. Exit");
            System.out.print("Choose an option: ");
            int choice = scanner.nextInt();
            scanner.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    System.out.println("\n--- Market Data ---");
                    for (String symbol : market.keySet()) {
                        System.out.println(symbol + ": $" + market.get(symbol));
                    }
                    break;

                case 2:
                    System.out.print("Enter stock symbol to buy: ");
                    String buySymbol = scanner.nextLine().toUpperCase();
                    if (market.containsKey(buySymbol)) {
                        System.out.print("Enter quantity to buy: ");
                        int qty = scanner.nextInt();
```

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double cost = qty * market.get(buySymbol);
if (cost <= balance) {
    balance -= cost;
    portfolio.put(buySymbol, portfolio.getDefault(buySymbol, 0) + qty);
    System.out.println("Bought " + qty + " shares of " + buySymbol);
} else {
    System.out.println("Insufficient balance!");
}
} else {
    System.out.println("Invalid stock symbol.");
}
break;

```

case 3:

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System.out.print("Enter stock symbol to sell: ");
String sellSymbol = scanner.nextLine().toUpperCase();
if (portfolio.containsKey(sellSymbol)) {
    System.out.print("Enter quantity to sell: ");
    int qty = scanner.nextInt();
    int owned = portfolio.get(sellSymbol);
    if (qty <= owned) {
        double value = qty * market.get(sellSymbol);
        balance += value;
        portfolio.put(sellSymbol, owned - qty);
        if (portfolio.get(sellSymbol) == 0) {
            portfolio.remove(sellSymbol);
        }
        System.out.println("Sold " + qty + " shares of " + sellSymbol);
    } else {
        System.out.println("You don't have that many shares.");
    }
} else {
    System.out.println("You don't own this stock.");
}
break;

```

case 4:

```

System.out.println("\n--- Portfolio ---");
double totalValue = 0;
for (String symbol : portfolio.keySet()) {
    int qty = portfolio.get(symbol);
    double value = qty * market.get(symbol);
    totalValue += value;
    System.out.println(symbol + ": " + qty + " shares ($" + value + ")");
}
System.out.println("Cash Balance: $" + balance);
System.out.println("Total Portfolio Value: $" + (balance + totalValue));
break;

```

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        case 5:
            running = false;
            System.out.println("Exiting platform.");
            break;

        default:
            System.out.println("Invalid option.");
    }
}

scanner.close();
}
```